2.05 Complex Numbers: Roots of Unity

Worked Solutions

1 a)

$$\begin{aligned} e^{\frac{\pi}{2}i} &= \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \\ &= i \end{aligned}$$

1 b)

$$\begin{aligned} e^{\frac{3\pi}{4}i} &= \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4} \\ &= -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \end{aligned}$$

1 c)

$$\begin{aligned} e^{\pi i} &= \cos\pi + i\sin\pi \\ &= -1 \end{aligned}$$

1 d)

$$\begin{aligned} e^{\frac{3\pi}{2}i} &= \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} \\ &= -i \end{aligned}$$

1 e)

$$\begin{aligned} e^{\frac{7\pi}{6}i} &= \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6} \\ &= -\frac{\sqrt{3}}{2} - \frac{1}{2}i \end{aligned}$$

1 f)

$$\begin{aligned} e^{2\pi i} &= \cos 2\pi + i\sin 2\pi \\ &= 1 \end{aligned}$$

2 a)

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

But $r^3 = 1 \Rightarrow r = 1$ Since $3\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{3}$ When $k = 0$:

$$z_0 = \mathrm{e}^0 = 1$$

When $k = 1$:

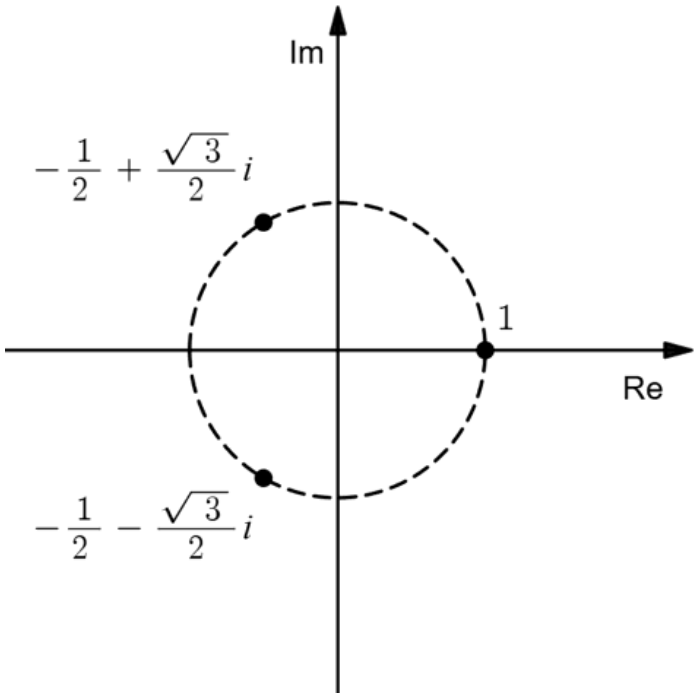
$$z_1 = \mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$$

When $k = 2$:

$$z_2 = \mathrm{e}^{\frac{4\pi}{3}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$$

$$\therefore z = 1, \quad z = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}, \quad z = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$$

2 b)



2 c)

Equilateral triangle.

2 d)

$$z_0 + z_1 + z_2 = 1 - \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} - \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} = 0$$

2 e)

Consider a triangle formed with two of the points and the centre:

$$\frac{1}{2} \times 1 \times 1 \times \sin \frac{2\pi}{3} = \frac{\sqrt{3}}{4}$$

Three triangles form the full shape:

$$\frac{3\sqrt{3}}{4} \text{ units}^2$$

3 a)

$$z^5 = r^5(\cos\theta + \mathrm{i}\sin\theta)^5 = r^5(\cos 5\theta + \mathrm{i}\sin 5\theta)$$

But $r^5 = 1 \Rightarrow r = 1$ Since $5\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{5}$ When $k = 0$:

$$z_0 = \mathrm{e}^0$$

When $k = 1$:

$$z_1 = \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}$$

When $k = 2$:

$$z_2 = \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}$$

When $k = 3$:

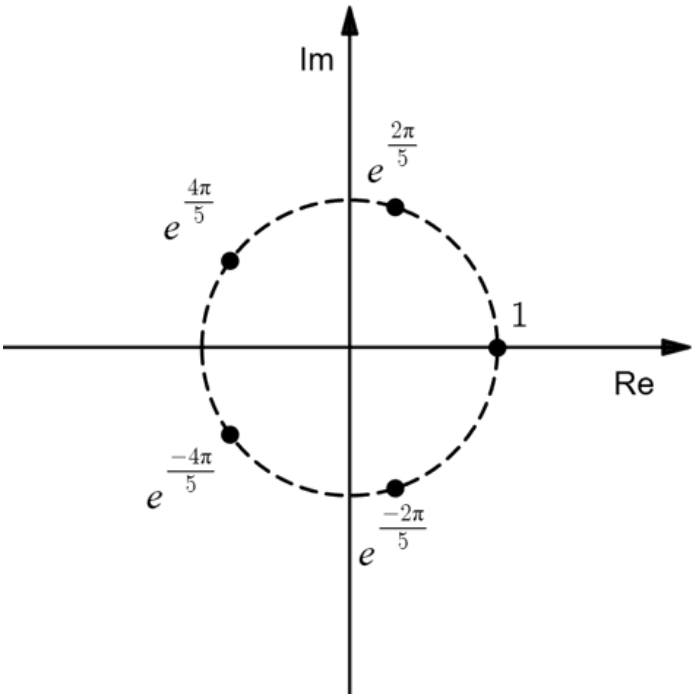
$$z_3 = \mathrm{e}^{\frac{6\pi}{5}\mathrm{i}} = \mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}$$

When $k = 4$:

$$z_4 = \mathrm{e}^{\frac{8\pi}{5}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}}$$

$$\therefore z = \mathrm{e}^0, \quad z = \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}}, \quad z = \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}}, \quad z = \mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}}, \quad z = \mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}}$$

3 b)



3 c)

Regular pentagon.

3 d)

$$z_0 + z_1 + z_2 + z_3 + z_4 + z_5 = \mathrm{e}^0 + \mathrm{e}^{\frac{2\pi}{5}\mathrm{i}} + \mathrm{e}^{\frac{4\pi}{5}\mathrm{i}} + \mathrm{e}^{-\frac{4\pi}{5}\mathrm{i}} + \mathrm{e}^{-\frac{2\pi}{5}\mathrm{i}} = 0$$

3 e)

Consider a triangle formed with two of the points and the centre:

$$\frac{1}{2} \times 1 \times 1 \times \sin\frac{2\pi}{5} = \frac{1}{2}\sin\frac{2\pi}{5}$$

Five triangles form the full shape:

$$\frac{5}{2}\sin\frac{2\pi}{5} = 2.38 \text{ units}^2 \text{ (3 SF)}$$

4 a)

$$z^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 5\theta + \mathrm{i}\sin 5\theta)$$

But $r^6 = 1 \Rightarrow r = 1$ Since $6\theta = 0 + 2k\pi$ for integer k: $\theta = \frac{2k\pi}{6}$ When $k = 0$:

$$z_0 = \mathrm{e}^0 = 1$$

When $k = 1$:

$$z_1 = \mathrm{e}^{\frac{2\pi}{6}\mathrm{i}} = \mathrm{e}^{\frac{\pi}{3}\mathrm{i}} = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$$

When $k = 2$:

$$z_2 = \mathrm{e}^{\frac{4\pi}{6}\mathrm{i}} = \mathrm{e}^{\frac{2\pi}{3}\mathrm{i}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$$

When $k = 3$:

$$z_3 = \mathrm{e}^{\frac{6\pi}{6}\mathrm{i}} = \mathrm{e}^{\pi\mathrm{i}} = -1$$

When $k = 4$:

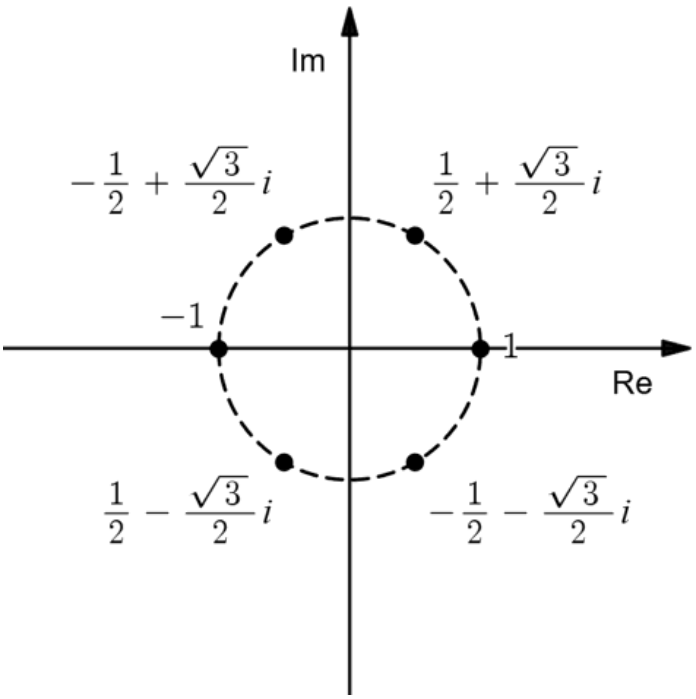
$$z_4 = \mathrm{e}^{\frac{8\pi}{6}\mathrm{i}} = \mathrm{e}^{-\frac{\pi}{3}\mathrm{i}} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$$

When $k = 5$:

$$z_5 = \mathrm{e}^{\frac{10\pi}{6}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{3}\mathrm{i}} = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$$

$$\therefore z = 1, \quad z = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}, \quad z = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}, \quad z = -1, \quad z = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}, \quad z = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$$

4 b)



4 c)

$$z_0+z_1+z_2+z_3+z_4+z_5=1+\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}-1+\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}=0$$

4 d)

Consider a triangle formed with two of the points and the centre:

$$\frac{1}{2}\times 1\times 1\times \sin\frac{\pi}{3}=\frac{\sqrt{3}}{4}$$

Six triangles form the full shape:

$$6\times\frac{\sqrt{3}}{4}=\frac{3\sqrt{3}}{2}\text{ units}^2\text{ (3 SF)}$$

5 a)

$$z^7=r^7(\cos\theta+\mathrm{i}\sin\theta)^7=r^7(\cos 7\theta+\mathrm{i}\sin 7\theta)$$

But $r^7=1\Rightarrow r=1$ Since $7\theta=0+2k\pi$ for integer k: $\theta=\frac{2k\pi}{7}$ When $k=0$:

$$z_0=\mathrm{e}^0$$

When $k=1$:

$$z_1=\mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}$$

When $k=2$:

$$z_2=\mathrm{e}^{\frac{4\pi}{7}\mathrm{i}}$$

When $k=3$:

$$z_3=\mathrm{e}^{\frac{6\pi}{7}\mathrm{i}}$$

When $k = 4$:

$$z_4 = \mathrm{e}^{\frac{8\pi}{7}\mathrm{i}} = \mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}}$$

When $k = 5$:

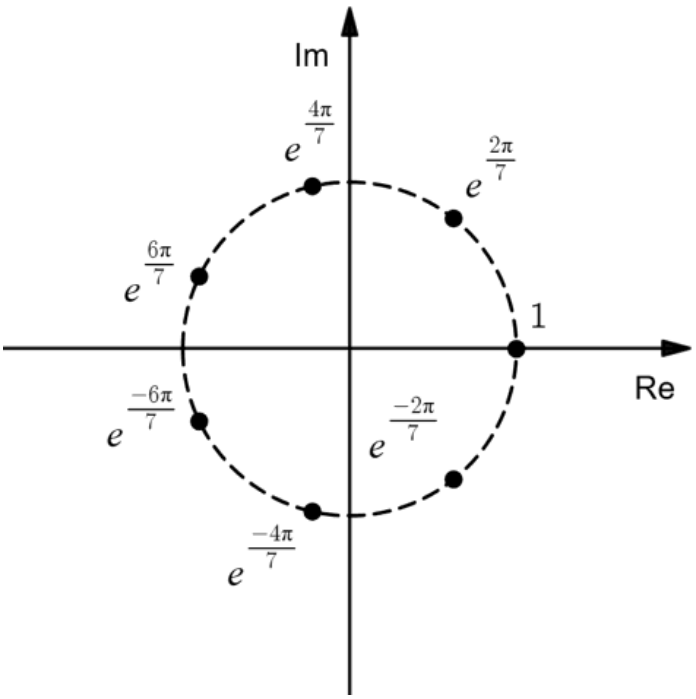
$$z_5 = \mathrm{e}^{\frac{10\pi}{7}\mathrm{i}} = \mathrm{e}^{-\frac{4\pi}{7}\mathrm{i}}$$

When $k = 6$:

$$z_6 = \mathrm{e}^{\frac{12\pi}{7}\mathrm{i}} = \mathrm{e}^{-\frac{2\pi}{7}\mathrm{i}}$$

$$\therefore z = \mathrm{e}^{0}, \quad z = \mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}, \quad z = \mathrm{e}^{\frac{4\pi}{7}\mathrm{i}}, \quad z = \mathrm{e}^{\frac{6\pi}{7}\mathrm{i}}, \quad z = \mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}}, \quad z = \mathrm{e}^{-\frac{4\pi}{7}\mathrm{i}}, \quad z = \mathrm{e}^{-\frac{2\pi}{7}\mathrm{i}}$$

5 b)



5 c)

$$\mathrm{e}^{0} + \mathrm{e}^{\frac{2\pi}{7}\mathrm{i}} + \mathrm{e}^{\frac{4\pi}{7}\mathrm{i}} + \mathrm{e}^{\frac{6\pi}{7}\mathrm{i}} + \mathrm{e}^{-\frac{6\pi}{7}\mathrm{i}} + \mathrm{e}^{-\frac{4\pi}{7}\mathrm{i}} + \mathrm{e}^{-\frac{2\pi}{7}\mathrm{i}} = 0$$

5 d)

Consider a triangle formed with two of the points and the centre:

$$\frac{1}{2} \times 1 \times 1 \times \sin\frac{2\pi}{7} = \frac{1}{2}\sin\frac{2\pi}{7}$$

Seven triangles form the full shape:

$$\frac{7}{2}\sin\frac{2\pi}{7} = 2.74 \text{ units}^2 \text{ (3 SF)}$$

6 a)

Consider a triangle formed with two of the points and the centre:

$$\frac{1}{2} \times 1 \times 1 \times \sin\frac{2\pi}{n} = \frac{1}{2}\sin\frac{2\pi}{n}$$

n triangles form the full shape:

$$\frac{n}{2}\sin\frac{2\pi}{n} \text{ units}^2$$

6 b)

As $n \to \infty$ the shape formed gets closer to a circle with radius 1

$$\therefore \pi \times 1^2 = \pi \text{ units}^2$$

7)

Let two consecutive roots of unity be $z_1 = \mathrm{e}^{\mathrm{i}\theta}$ and $z_2 = \mathrm{e}^{\mathrm{i}(\theta+2\pi/n)}$. The angle between them is $\theta = \frac{2\pi}{n}$. The distance between them is the modulus of their difference:

$$\begin{aligned} |z_2 - z_1| &= |\mathrm{e}^{\mathrm{i}(\theta+2\pi/n)} - \mathrm{e}^{\mathrm{i}\theta}| \\ &= |\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{2\pi\mathrm{i}/n} - \mathrm{e}^{\mathrm{i}\theta}| \\ &= |\mathrm{e}^{\mathrm{i}(\theta+\pi/n)}|\,|\mathrm{e}^{\pi\mathrm{i}/n} - \mathrm{e}^{-\pi\mathrm{i}/n}| \\ &= |\mathrm{e}^{\pi\mathrm{i}/n} - \mathrm{e}^{-\pi\mathrm{i}/n}| \end{aligned}$$

Since $\mathrm{e}^{\mathrm{i}\alpha} - \mathrm{e}^{-\mathrm{i}\alpha} = 2\mathrm{i}\sin\alpha$,

$$|\mathrm{e}^{\pi\mathrm{i}/n} - \mathrm{e}^{-\pi\mathrm{i}/n}| = |2\mathrm{i}\sin(\pi/n)| = 2|\sin(\pi/n)|.$$

For a distance to exist, $n \geq 2$, so $0 < \frac{\pi}{n} \leq \frac{\pi}{2}$ and $\sin\frac{\pi}{n} \geq 0$. Hence

$$|z_2 - z_1| = 2\sin\frac{\pi}{n}.$$

Alternatively, using the cosine rule, the distance is

$$\sqrt{1^2 + 1^2 - 2 \times 1 \times 1 \times \cos(2\pi/n)} = \sqrt{2 - 2\cos(2\pi/n)}.$$

Since $\cos(2A) = 1 - 2\sin^2 A$, this is

$$\sqrt{4\sin^2(\pi/n)} = \sin\frac{\pi}{n} \quad \text{as required.}$$

8 a)

First point: $2\mathrm{e}^0$. Other points: $2\mathrm{e}^{2\pi\mathrm{i}/3}$, $2\mathrm{e}^{-2\pi\mathrm{i}/3}$.

$$-1 + \sqrt{3}\mathrm{i}, \quad -1 - \sqrt{3}\mathrm{i}$$

$$(-1, \sqrt{3}) \quad \text{and} \quad (-1, -\sqrt{3})$$

8 b)

Three triangles of angle $\frac{2\pi}{3}$:

$$3 \times \left(\frac{1}{2} \times 2 \times 2 \times \sin\frac{2\pi}{3}\right) = 3\sqrt{3} \text{ units}^2.$$

9 a)

First point: $3\mathrm{e}^0$. Other points: $3\mathrm{e}^{2\pi\mathrm{i}/6}$, $3\mathrm{e}^{4\pi\mathrm{i}/6}$, $3\mathrm{e}^{\pi\mathrm{i}}$, $3\mathrm{e}^{-4\pi\mathrm{i}/6}$, $3\mathrm{e}^{-2\pi\mathrm{i}/6}$.

$$\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}, \quad -\frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}, \quad -3, \quad -\frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}, \quad \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}$$

$$\left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right), \left(-\frac{3}{2}, \frac{3\sqrt{3}}{2}\right), (-3, 0), \left(-\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right), \left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}\mathrm{i}\right)$$

9 b)

Six triangles of angle $\frac{\pi}{3}$:

$$6 \times \left(\frac{1}{2} \times 3 \times 3 \times \sin \frac{\pi}{3} \right) = \frac{27\sqrt{3}}{2} \text{ units}^2.$$

10)

Let the root be $z = \mathrm{e}^{\mathrm{i}\theta}$, so its complex conjugate is $z^* = \mathrm{e}^{-\mathrm{i}\theta}$. The nth power of the conjugate is

$$(z^*)^n = (\mathrm{e}^{-\mathrm{i}\theta})^n = \mathrm{e}^{-\mathrm{i}n\theta}.$$

Since $z^n = \mathrm{e}^{\mathrm{i}n\theta} = 1$,

$$(z^*)^n = \mathrm{e}^{-\mathrm{i}n\theta} = (\mathrm{e}^{\mathrm{i}n\theta})^* = (1)^* = 1.$$

So z^* is also an nth root of unity.

11)

Assume the first root is at 1, so if there is only one root then the result is true. Let $\omega = \mathrm{e}^{2\pi\mathrm{i}/n}$, so the sum of the roots is

$$S_n = 1 + \omega + \omega^2 + \cdots + \omega^{n-1}.$$

This is a geometric series with $a = 1$, $r = \omega$ and n terms:

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{1(1 - \omega^n)}{1 - \omega}.$$

But $\omega^n = 1$, so

$$S_n = \frac{1(1 - 1)}{1 - \omega} = 0.$$

12 a)

$$\left(\cos \frac{4\pi}{5} + \mathrm{i} \sin \frac{4\pi}{5} \right)^5 = \cos \left(5 \times \frac{4\pi}{5} \right) + \mathrm{i} \sin \left(5 \times \frac{4\pi}{5} \right) = 1 + 0\mathrm{i} = 1,$$

so $\cos \frac{4\pi}{5} + \mathrm{i} \sin \frac{4\pi}{5}$ is a solution to $z^5 = 1$.

12 b)

$$(\mathrm{e}^{2\pi\mathrm{i}/7})^7 = \mathrm{e}^{7 \times 2\pi\mathrm{i}/7} = \mathrm{e}^{2\pi\mathrm{i}} = \mathrm{e}^0 = 1,$$

so $\mathrm{e}^{2\pi\mathrm{i}/7}$ is a root of $z^7 = 1$.

13 a)

$$z^5 = r^5(\cos\theta + \mathrm{i}\sin\theta)^5 = r^5(\cos 5\theta + \mathrm{i}\sin 5\theta)$$

But $r^5 = 1 \Rightarrow r = 1$

Since $5\theta = 0 + 2k\pi$ for integer k, $\theta = \frac{2k\pi}{5}$.

$$k = 0: \quad z_0 = \cos 0 + \mathrm{i}\sin 0$$

$$k = 1: \quad z_1 = \cos \frac{2\pi}{5} + \mathrm{i}\sin \frac{2\pi}{5}$$

$$k=2: \quad z_2=\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}$$
$$k=3: \quad z_3=\cos\frac{6\pi}{5}+\mathrm{i}\sin\frac{6\pi}{5}=\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)$$
$$k=4: \quad z_4=\cos\frac{8\pi}{5}+\mathrm{i}\sin\frac{8\pi}{5}=\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right)$$

$$\therefore z=\cos 0+\mathrm{i}\sin 0,\ z=\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5},$$
$$z=\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5},\ z=\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right),$$
$$z=\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right).$$

13 b)

$$\cos 0+\mathrm{i}\sin 0+\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}+\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5}$$
$$+\cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right)+\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right)=0$$

Consider the real parts:

$$1+\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}+\cos\left(-\frac{4\pi}{5}\right)+\cos\left(-\frac{2\pi}{5}\right)=0.$$

Since $\cos\frac{2\pi}{5}=\cos(-\frac{2\pi}{5})$ and $\cos\frac{4\pi}{5}=\cos(-\frac{4\pi}{5})$,

$$1+2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=0$$
$$\Rightarrow 2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=-1$$
$$\therefore \cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}=-\frac{1}{2} \text{ as required.}$$

14 a)

$$z^7=r^7(\cos\theta+\mathrm{i}\sin\theta)^7=r^7(\cos 7\theta+\mathrm{i}\sin 7\theta)$$
$$\text{But } r^7=1\Rightarrow r=1$$

Since $7\theta=0+2k\pi$ for integer k, $\theta=\frac{2k\pi}{7}$.

$$k=0: z_0=\cos 0+\mathrm{i}\sin 0$$
$$k=1: z_1=\cos\frac{2\pi}{7}+\mathrm{i}\sin\frac{2\pi}{7}$$
$$k=2: z_2=\cos\frac{4\pi}{7}+\mathrm{i}\sin\frac{4\pi}{7}$$
$$k=3: z_3=\cos\frac{6\pi}{7}+\mathrm{i}\sin\frac{6\pi}{7}$$
$$k=4: z_4=\cos\frac{8\pi}{7}+\mathrm{i}\sin\frac{8\pi}{7}=\cos\left(-\frac{6\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{6\pi}{7}\right)$$

$$k=5: z_5=\cos\frac{10\pi}{7}+\mathrm{i}\sin\frac{10\pi}{7}=\cos\left(-\frac{4\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{7}\right)$$
$$k=6: z_6=\cos\frac{12\pi}{7}+\mathrm{i}\sin\frac{12\pi}{7}=\cos\left(-\frac{2\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{7}\right)$$

$$\therefore z=\cos 0+\mathrm{i}\sin 0,\ z=\cos\frac{2\pi}{7}+\mathrm{i}\sin\frac{2\pi}{7},\ z=\cos\frac{4\pi}{7}+\mathrm{i}\sin\frac{4\pi}{7},\ z=\cos\frac{6\pi}{7}+\mathrm{i}\sin\frac{6\pi}{7},$$
$$z=\cos\left(-\frac{6\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{6\pi}{7}\right),\ z=\cos\left(-\frac{4\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{7}\right),\ z=\cos\left(-\frac{2\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{7}\right).$$

14 b)

$$\cos 0+\mathrm{i}\sin 0+\cos\frac{2\pi}{7}+\mathrm{i}\sin\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\mathrm{i}\sin\frac{4\pi}{7}+\cos\frac{6\pi}{7}+\mathrm{i}\sin\frac{6\pi}{7}$$
$$+\cos\left(-\frac{6\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{6\pi}{7}\right)+\cos\left(-\frac{4\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{7}\right)+\cos\left(-\frac{2\pi}{7}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{7}\right)=0$$

Consider the real parts:

$$1+\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}+\cos\left(-\frac{6\pi}{7}\right)+\cos\left(-\frac{4\pi}{7}\right)+\cos\left(-\frac{2\pi}{7}\right)=0.$$

Since $\cos\frac{2\pi}{7}=\cos(-\frac{2\pi}{7})$, $\cos\frac{4\pi}{7}=\cos(-\frac{4\pi}{7})$ and $\cos\frac{6\pi}{7}=\cos(-\frac{6\pi}{7})$,

$$1+2\cos\frac{2\pi}{7}+2\cos\frac{4\pi}{7}+2\cos\frac{6\pi}{7}=0$$
$$\Rightarrow 2\cos\frac{2\pi}{7}+2\cos\frac{4\pi}{7}+2\cos\frac{6\pi}{7}=-1$$
$$\therefore \cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}=-\frac{1}{2}\text{ as required.}$$

15)

The five roots of unity are

$$\cos 0+\mathrm{i}\sin 0,\ \cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5},$$
$$\cos\frac{4\pi}{5}+\mathrm{i}\sin\frac{4\pi}{5},\ \cos\left(-\frac{4\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{4\pi}{5}\right),$$
$$\cos\left(-\frac{2\pi}{5}\right)+\mathrm{i}\sin\left(-\frac{2\pi}{5}\right).$$

Their sum is zero. Consider the real parts:

$$1+\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}+\cos\left(-\frac{4\pi}{5}\right)+\cos\left(-\frac{2\pi}{5}\right)=0.$$

Since $\cos a=\cos(-a)$,

$$1+2\cos\frac{2\pi}{5}+2\cos\frac{4\pi}{5}=0\Rightarrow\cos\frac{2\pi}{5}+\cos\frac{4\pi}{5}=-\frac{1}{2}\Rightarrow-\cos\frac{2\pi}{5}-\cos\frac{4\pi}{5}=\frac{1}{2}.$$

However, $\cos(\pi-\alpha)=-\cos\alpha$. Since $\cos\frac{3\pi}{5}=\cos(\pi-\frac{2\pi}{5})=-\cos\frac{2\pi}{5}$ and $\cos\frac{\pi}{5}=\cos(\pi-\frac{4\pi}{5})=-\cos\frac{4\pi}{5}$,

$$\therefore \cos\frac{3\pi}{5}+\cos\frac{\pi}{5}=\frac{1}{2}\text{ as required.}$$

16 a)

Since $3\theta = 0 + 2k\pi$ for integer k, $\theta = \frac{2k\pi}{3}$.

$$\begin{aligned} k = 0: &\quad z_0 = \mathrm{e}^0 = 1 \\ k = 1: &\quad z_1 = \mathrm{e}^{2\pi\mathrm{i}/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} \\ k = 2: &\quad z_2 = \mathrm{e}^{4\pi\mathrm{i}/3} = \mathrm{e}^{-2\pi\mathrm{i}/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} \end{aligned}$$

$$\therefore 1,\ -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i},\ -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}.$$

$$\begin{aligned} \left(z - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z - \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)\right) &= z^2 + \left(\frac{1}{2} + \frac{1}{2}\right)z + \left(\frac{1}{4} + \frac{3}{4}\right) \\ &= z^2 + z + 1. \end{aligned}$$

So

$$(z-1)(z^2 + z + 1) = z^3 + z^2 + z - z^2 - z - 1 = z^3 - 1.$$

16 b)

From part b), the three cube roots of unity are $1, -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}, -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$. The midpoints between each pair on the Argand diagram are

$$\begin{aligned} 1 \text{ and } -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}: &\quad \frac{1}{4} + \frac{\sqrt{3}}{4}\mathrm{i}, \\ -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} \text{ and } -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}: &\quad -\frac{1}{2}, \\ -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} \text{ and } 1: &\quad \frac{1}{4} - \frac{\sqrt{3}}{4}\mathrm{i}. \end{aligned}$$

$$\begin{aligned} \left(z - (\tfrac{1}{4} + \tfrac{\sqrt{3}}{4}\mathrm{i})\right)\left(z - (\tfrac{1}{4} - \tfrac{\sqrt{3}}{4}\mathrm{i})\right) &= z^2 - \tfrac{1}{2}z + \tfrac{1}{4} \\ (z^2 - \tfrac{1}{2}z + \tfrac{1}{4})(z + \tfrac{1}{2}) &= z^3 - \tfrac{1}{2}z^2 + \tfrac{1}{4}z + \tfrac{1}{2}z^2 - \tfrac{1}{4}z + \tfrac{1}{8} \\ &= z^3 + \tfrac{1}{8} \end{aligned}$$

So since $z^3 + \frac{1}{8} = 0$, $8z^3 + 1 = 0$.

$$\therefore 8z^3 + 1.$$

17 a)

$1 + z + z^2 + z^3 + z^4 + z^5$ is a geometric series, so we are solving

$$\frac{z^6 - 1}{z - 1} = 0.$$

Since we use this quotient, $z \neq 1$. Thus the solutions are the sixth roots of unity. They are found from $6\theta = 0 + 2k\pi$ for integer k, so $\theta = \frac{2k\pi}{6}$:

$$\begin{aligned} k = 0 &: z_0 = 1 \\ k = 1 &: z_1 = \mathrm{e}^{2\pi\mathrm{i}/6} = \mathrm{e}^{\pi\mathrm{i}/3} = \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i} \\ k = 2 &: z_2 = \mathrm{e}^{4\pi\mathrm{i}/6} = \mathrm{e}^{2\pi\mathrm{i}/3} = -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i} \end{aligned}$$

$$k = 3 : z_3 = \mathrm{e}^{6\pi\mathrm{i}/6} = \mathrm{e}^{\pi\mathrm{i}} = -1$$
$$k = 4 : z_4 = \mathrm{e}^{8\pi\mathrm{i}/6} = \mathrm{e}^{-\pi\mathrm{i}/3} = \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i}$$
$$k = 5 : z_5 = \mathrm{e}^{8\pi\mathrm{i}/6} = \mathrm{e}^{-2\pi\mathrm{i}/3} = -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i}$$

$$\therefore z = -1,\ z = -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i},\ z = \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i},\ z = -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i},\ z = \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i}.$$

17 b)

$$\left(z - (-\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i})\right)\left(z - (-\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i})\right) = z^2 + z + 1$$
$$\left(z - (\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i})\right)\left(z - (\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i})\right) = z^2 - z + 1$$

$$\therefore 1 + z + z^2 + z^3 + z^4 + z^5 = (z+1)(z^2 + z + 1)(z^2 - z + 1).$$

18 a)

$1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7$ is a geometric series, so we solve

$$\frac{z^8 - 1}{z - 1} = 0,$$

where $z \neq 1$. The solutions are the eighth roots of unity. From $8\theta = 0 + 2k\pi$ for integer k, $\theta = \frac{2k\pi}{4}$:

$$k = 0 : z_0 = 1$$
$$k = 1 : z_1 = \mathrm{e}^{2\pi\mathrm{i}/8} = \mathrm{e}^{\pi\mathrm{i}/4} = \tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{2}\mathrm{i}$$
$$k = 2 : z_2 = \mathrm{e}^{4\pi\mathrm{i}/8} = \mathrm{e}^{\pi\mathrm{i}/2} = \mathrm{i}$$
$$k = 3 : z_3 = \mathrm{e}^{6\pi\mathrm{i}/8} = \mathrm{e}^{3\pi\mathrm{i}/4} = -\tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{2}\mathrm{i}$$
$$k = 4 : z_4 = \mathrm{e}^{8\pi\mathrm{i}/8} = \mathrm{e}^{\pi\mathrm{i}} = -1$$
$$k = 5 : z_5 = \mathrm{e}^{10\pi\mathrm{i}/8} = \mathrm{e}^{-3\pi\mathrm{i}/4} = -\tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2}\mathrm{i}$$
$$k = 6 : z_6 = \mathrm{e}^{12\pi\mathrm{i}/8} = \mathrm{e}^{-\pi\mathrm{i}/3} = -\mathrm{i}$$
$$k = 7 : z_7 = \mathrm{e}^{14\pi\mathrm{i}/8} = \mathrm{e}^{-\pi\mathrm{i}/2} = \tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2}\mathrm{i}$$

$$\therefore z = -1,\ z = -\tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2}\mathrm{i},\ z = -\mathrm{i},\ z = \tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2}\mathrm{i},\ z = 1,\ z = \tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{2}\mathrm{i},\ z = \mathrm{i},\ z = -\tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{2}\mathrm{i}.$$

18 b)

$$\left(z - (\tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{2}\mathrm{i})\right)\left(z - (\tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2}\mathrm{i})\right) = z^2 - \sqrt{2}z + 1$$
$$(z - \mathrm{i})(z - (-\mathrm{i})) = z^2 + 1$$
$$\left(z - (-\tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{2}\mathrm{i})\right)\left(z - (-\tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2}\mathrm{i})\right) = z^2 + \sqrt{2}z + 1$$

And

$$(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1) = z^4 + (\sqrt{2} - \sqrt{2})z^3 + (1 + 1 - 2)z^2 + (-\sqrt{2} + \sqrt{2})z + 1$$
$$= z^4 + 1.$$

$$\therefore 1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7 = (z+1)(z^2+1)(z^4+1).$$

19 a)

Roots are $1, \omega, \omega^2$, so $\omega^3 = 1$.

19 b)

Roots are $1, \omega, \omega^2$, so $\omega^5 = \omega^2$.

19 c)

Roots are $1, \omega, \omega^2$, so $\omega^8 = \omega^5 = \omega^2$ and $\omega^6 = \omega^3 = 1$.

$$\therefore \omega^8 - \omega^5 + \omega^6 = \omega^2 - \omega^2 + 1 = 1.$$

19 d)

Roots are $1, \omega, \omega^2$, so

$$\frac{1}{\omega} = \omega^{-1} = \omega^2.$$

20 a)

$\omega = \omega^1 = \mathrm{e}^{2\pi\mathrm{i}/n}$, so $\omega^* = \mathrm{e}^{-2\pi\mathrm{i}/n}$. But since they are roots of unity,

$$\omega^* = \mathrm{e}^{-2\pi\mathrm{i}/n} = \mathrm{e}^{2(n-1)\pi\mathrm{i}/n} = \omega^{n-1}.$$

Alternatively, $\omega^n = 1$, so $\omega \times \omega^* = 1$ and $\omega \times \omega^* = \omega^n$. Hence $\omega^* = \omega^{n-1}$.

20 b)

$$(\omega^k)^* = (\omega^*)^k = (\omega^{n-1})^k = \omega^{k(n-1)}.$$

Since it has n roots, this gives $(\omega^k)^* = \omega^{n-k}$. Alternatively, $(\omega^k)^n = 1$, so $\omega^k \times (\omega^k)^* = 1$ and $\omega^k \times (\omega^k)^* = (\omega^k)^n$. Hence $(\omega^k)^* = \omega^{k(n-1)}$.

21 a)

There are six roots of unity, so $1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 = 0$.

$$\mathrm{Re}(\omega + \omega^2 + \omega^3 + \omega^4 + \omega^5) = \mathrm{Re}(-1) = -1.$$

21 b)

Since $\omega = \mathrm{e}^{2\pi\mathrm{i}/6}$, $\omega^2 = \mathrm{e}^{2\pi\mathrm{i}/3}$ and $\omega^4 = \mathrm{e}^{-2\pi\mathrm{i}/3}$.

$$\begin{aligned}\mathrm{Im}(\omega^2 + \omega^4) &= \mathrm{Im}(\mathrm{e}^{2\pi\mathrm{i}/3} + \mathrm{e}^{-2\pi\mathrm{i}/3}) \\ &= \sin\frac{2\pi}{3} + \sin\left(-\frac{2\pi}{3}\right) = 0.\end{aligned}$$

22 a)

Since $\omega = \mathrm{e}^{\pi\mathrm{i}/4}$, $\omega^2 = \mathrm{e}^{2\pi\mathrm{i}/4}$.

$$\mathrm{Re}(\omega^2 + \omega) = \cos\frac{\pi}{2} + \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}.$$

22 b)

$$\omega^4 = 0$$
$$\mathrm{Im}(1+\omega^4) = \mathrm{Im}(1+0) = 0.$$

23)

The four roots of unity are $1, \omega = \mathrm{i}, \omega^2 = -1, \omega^3 = -\mathrm{i}$.

$$\begin{aligned}(1-\mathrm{i})(1+1)(1+\mathrm{i}) &= 2(1+\mathrm{i})(1-\mathrm{i}) \\ &= 2(1+1) = 4 \quad \text{as required.}\end{aligned}$$

24 a)

$$a = 1, \quad r = x, \quad n = n.$$
$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1(1-x^n)}{1-x} = \frac{1-x^n}{1-x}.$$

24 b)

Since

$$z^n - 1 = (z-1)(z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}),$$

for $z \neq 1$ divide by $z-1$ to get

$$\frac{z^n-1}{z-1} = (z-\omega)(z-\omega^2)\cdots(z-\omega^{n-1}),$$

which from part a) is $1 + z + z^2 + \cdots + z^{n-1}$. Both sides are equal for all $z \neq 1$ and define the same polynomial for $z \neq 1$, so take the limit as $z \to 1$. The sum gives $1+1+1+\cdots+1 = n$, while the product gives $(1-\omega)(1-\omega^2)\cdots(1-\omega^{n-1})$. Thus

$$\prod_{k=1}^{n-1}(1-\omega^k) = n \quad \text{as required.}$$

25 a)

Since $|z|^2 = zz^*$, let $z = z_1 + z_2$. Then

$$|z_1+z_2|^2 = (z_1+z_2)(z_1+z_2)^*.$$

Since conjugation distributes over addition,

$$|z_1+z_2|^2 = (z_1+z_2)(z_1^*+z_2^*).$$

25 b)

$$\begin{aligned}|\omega-\omega^2|^2 &= (\omega-\omega^2)(\omega^*-(\omega^2)^*)\\ &= (\omega-\omega^2)(\omega^*-(\omega^*)^2)\\ &= \omega\omega^*-\omega(\omega^*)^2-\omega^2\omega^*+\omega^2(\omega^*)^2.\end{aligned}$$

Since $|\omega|^2=\omega\omega^*=1$,

$$=1-\omega(\omega^*)^2-\omega^2\omega^*+1=2-(\omega(\omega^*)^2+\omega^2\omega^*).$$

Since ω is a cube root of unity, $\omega^2\omega^*=\frac{\omega^2}{\omega}=\omega$ and $\omega(\omega^*)^2=\frac{\omega}{\omega^2}=\frac{1}{\omega}=\omega^*$.

$$\therefore |\omega-\omega^2|^2=2-(\omega+\omega^*)\quad\text{as required.}$$

Alternatively, test the three cube roots. For $\omega=1$:

$$\text{LHS: } |1-1^2|^2=0,\qquad \text{RHS: } 2-(1+1^*)=0,$$

so LHS = RHS. For $\omega=-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}$:

$$\text{LHS: } \left|-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}-\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}\right)^2\right|^2=\left|-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}+\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}\right|^2=|\sqrt{3}\mathrm{i}|^2=3,$$

$$\text{RHS: } 2-\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}\right)=3,$$

so LHS = RHS. For $\omega=-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$:

$$\text{LHS: } \left|-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}-\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}\right)^2\right|^2=\left|-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}+\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}\right|^2=|-\sqrt{3}\mathrm{i}|^2=3,$$

$$\text{RHS: } 2-\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}-\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}\right)=3,$$

so LHS = RHS. Hence $|\omega-\omega^2|^2=2-(\omega+\omega^*)$.

26 a)

Since $1+\omega+\omega^2=0$,

$$(1-\omega)(1-\omega^2)=1-(\omega+\omega^2)+\omega^3=1-(-1)+1=3.$$

26 b)

$$1-\omega^3=1-1=0,\qquad (1-\omega)(1-\omega^2)=3\text{ from part a).}$$
$$\therefore (1-\omega)(1-\omega^2)(1-\omega^3)=3\times 0=0.$$

26 c)

Since $1 + \omega + \omega^2 = 0$,

$$(1 + \omega)(1 + \omega^2) = (-\omega^2)(-\omega) = \omega^3 = 1.$$

26 d)

Since $\omega^3 = 1$ and $(1 + \omega)(1 + \omega^2) = 1$ from part c),

$$(1 + \omega)(1 + \omega^2)(1 + \omega^3) = 1 \times (1 + 1) = 2.$$

Alternatively, since $1 + \omega + \omega^2 = 0$,

$$(1 + \omega)(1 + \omega^2)(1 + \omega^3) = (-\omega^2)(-\omega) \times 2 = 2\omega^3 = 2.$$

27 a)

Since $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$,

$$\begin{aligned}(a - b\omega)(a - b\omega^2) &= a^2 - ab(\omega + \omega^2) + b^2\omega^3 \\ &= a^2 - ab(-1) + b^2 = a^2 + ab + b^2.\end{aligned}$$

27 b)

Since $\omega^3 = 1$ and part a) gives $(a - b\omega)(a - b\omega^2) = a^2 + ab + b^2$,

$$\begin{aligned}(a - b\omega)(a - b\omega^2)(a - b\omega^3) &= (a^2 + ab + b^2)(a - b) \\ &= a^3 - b^3.\end{aligned}$$

27 c)

Since $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$,

$$\begin{aligned}(a + b\omega)(a + b\omega^2) &= a^2 + ab(\omega + \omega^2) + b^2\omega^3 \\ &= a^2 - ab + b^2.\end{aligned}$$

27 d)

Since $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$,

$$(a + b\omega)(a + b\omega^2)(a + b\omega^3).$$

We saw $(a + b\omega)(a + b\omega^2)$ in part c), so

$$\begin{aligned}(a^2 - ab + b^2)(a + b\omega^3) &= (a^2 - ab + b^2)(a + b) \\ &= a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 \\ &= a^3 + b^3.\end{aligned}$$

28 a)

Four roots of unity give $1 + \omega + \omega^2 + \omega^3 = 0$.

$$\therefore \omega(1 + \omega + \omega^2) = \omega \times (-\omega^3) = -\omega^4 = -1.$$

28 b)

Four roots of unity give $1+\omega+\omega^2+\omega^3=0$.

$$\therefore \omega(1+\omega)(1+\omega^2)=\omega(1+\omega+\omega^2+\omega^3)=0.$$

28 c)

Four roots of unity give $1+\omega+\omega^2+\omega^3=0$.

$$\begin{aligned}\left(\omega+\frac{1}{\omega}\right)^2+2\left(\omega+\frac{1}{\omega}\right) &= \omega^2+2+\omega^{-2}+2\omega+2\omega^{-1}\\ &= \omega^2+2+\omega^2+2\omega+2\omega^3\\ &= 2\omega^2+2+2\omega+2\omega^3\\ &= 2(1+\omega+\omega^2+\omega^3)=0.\end{aligned}$$

Here $\omega^{-2}=\omega^2$ and $\omega^{-1}=\omega^3$.

29 a)

$$\omega+\frac{1}{\omega}+1=\omega+\omega^{-1}+1.$$

Since there are three roots of unity, $\omega^{-1}=\omega^2$.

$$=\omega+\omega^2+1=0.$$

29 b)

From part a), let the sum of the conjugates be x:

$$x=\omega+\frac{1}{\omega}, \qquad x+1=0, \qquad \therefore x=-1.$$

Since $\omega=\mathrm{e}^{2\pi\mathrm{i}/5}$ and $\omega^*=\frac{1}{\omega}$,

$$\begin{aligned}x &= \omega+\frac{1}{\omega}\\ &= \left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)+\left(\cos\frac{2\pi}{3}-\mathrm{i}\sin\frac{2\pi}{3}\right)\\ &= 2\cos\frac{2\pi}{3}.\end{aligned}$$

Thus $2\cos\frac{2\pi}{3}=-1$, so $\cos\frac{2\pi}{3}=-\frac{1}{2}$ as required.

30 a)

$$\begin{aligned}\left(\omega+\frac{1}{\omega}\right)^2+\left(\omega+\frac{1}{\omega}\right) &= \omega^2+2+\omega^{-2}+\omega+\omega^{-1}\\ &= \omega^2+2+\omega^3+\omega+\omega^4\\ &= 1+1+\omega+\omega^2+\omega^3+\omega^4\\ &= 1+0=1.\end{aligned}$$

Here, since there are five roots of unity, $\omega^{-2}=\omega^3$ and $\omega^{-1}=\omega^4$.

30 b)

From part a), let the sum of the conjugates be x:

$$x = \omega + \frac{1}{\omega}, \qquad x^2 + x = 1 \Rightarrow x^2 + x - 1 = 0, \qquad \therefore x = \frac{-1 \pm \sqrt{5}}{2}.$$

Since $\omega = \mathrm{e}^{2\pi \mathrm{i}/5}$ and $\omega^* = \frac{1}{\omega}$,

$$\begin{aligned} x &= \omega + \frac{1}{\omega} = \omega + \omega^4 \\ &= \left(\cos\frac{2\pi}{5} + \mathrm{i}\sin\frac{2\pi}{5}\right) + \left(\cos\frac{2\pi}{5} - \mathrm{i}\sin\frac{2\pi}{5}\right) \\ &= 2\cos\frac{2\pi}{5}. \end{aligned}$$

But $\omega = \mathrm{e}^{2\pi \mathrm{i}/5}$ lies in the first quadrant, so $2\cos\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{2}$. But we want $\cos\frac{4\pi}{5}$:

$$\begin{aligned} \cos\frac{4\pi}{5} &= \cos\left(2 \times \frac{2\pi}{5}\right) = 2\left(\cos\frac{2\pi}{5}\right)^2 - 1 \\ &= 2\left(\frac{-1+\sqrt{5}}{4}\right)^2 - 1 = \frac{-1-\sqrt{5}}{4}. \end{aligned}$$

Alternatively, from the same root sum and real parts,

$$1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0 \Rightarrow \cos\frac{2\pi}{5} + \cos\frac{4\pi}{5} = -\frac{1}{2}.$$

Using $\cos\alpha = -\cos(\alpha - \pi)$,

$$-\cos\left(\frac{2\pi}{5} - \pi\right) - \cos\left(\frac{4\pi}{5} - \pi\right) = -\frac{1}{2}$$

$$-\cos\left(-\frac{3\pi}{5}\right) - \cos\left(-\frac{\pi}{5}\right) = -\frac{1}{2}$$

$$\therefore \cos\frac{3\pi}{5} + \cos\frac{\pi}{5} = \frac{1}{2} \text{ as required.}$$